LLM Applications in Cybersecurity

Term Project Proposal

Phishing & Social Engineering Detection via Fine-Tuned LLM Pipelines

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Course: LLM Applications in Cybersecurity

Group Number: 140

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1. Project Overview

Phishing and social engineering attacks remain among the most prevalent and economically damaging cybersecurity threats worldwide. Traditional rule-based and signature-based detection systems are increasingly insufficient against modern, contextually sophisticated phishing campaigns that exploit natural language ambiguity to evade filters. This proposal outlines a Proof-of-Concept (PoC) system that harnesses the contextual understanding capabilities of Large Language Models (LLMs) — specifically instruction-tuned transformer architectures — to detect phishing and social engineering attempts across multiple communication channels including email, SMS, and messaging platforms.

Our system will leverage fine-tuning on curated, domain-specific datasets along with advanced prompt engineering to achieve detection precision that surpasses current state-of-the-art baselines. The proposal is organized around five core components: motivation and threat landscape analysis, dataset curation and availability, proposed LLM architecture and methodology, evaluation framework, and implementation roadmap.

1.1 Motivation

According to the 2024 Verizon Data Breach Investigations Report, over 36% of all data breaches involve phishing as an initial access vector. The financial losses attributable to phishing-related business email compromise (BEC) exceeded USD $2.9 billion in 2023 alone. Critically, attackers now deploy AI-generated spear-phishing content that is syntactically and semantically indistinguishable from legitimate correspondence, rendering keyword and n-gram based classifiers obsolete. An LLM-based detection pipeline that can reason over full message context, sender behavior patterns, and cross-channel signals represents a qualitative leap in defensive capability.

1.2 Learning Objectives Addressed

- Advanced Technical Application: Fine-tuning transformer models (DistilBERT, RoBERTa, LLaMA-3) on heterogeneous phishing datasets and constructing multi-stage prompt pipelines for contextual threat classification.
- Independent Ideation & Research: Designing a novel multi-modal, multi-channel phishing detection architecture that fuses email header analysis, URL reputation scoring, and LLM semantic classification into a unified detection pipeline.

2. Datasets & Available Resources

One of the primary strengths of the Phishing & Social Engineering Detection domain is the abundance of high-quality, publicly available datasets. The following table summarizes the key datasets that will be used in this project:

Dataset	Source	Size	Type	License
CEAS-08 Spam/Phishing Corpus	CEAS Conference	~39,000 emails	Email (raw)	Public
Enron Email Dataset	CMU / Kaggle	~500,000 emails	Email (ham)	Public
SpamAssassin Public Corpus	Apache	~6,000 emails	Email mixed	Apache 2.0
PhishTank URL Dataset	PhishTank.org	~1M+ URLs	URL labels	CC BY 2.5
ISCX-URL-2016	UNB CIC	~35,000 URLs	URL features	Research
SMS Spam Collection	UCI ML Repo	~5,574 SMS	SMS text	Public
Mendeley Phishing Email Dataset	Mendeley Data	~82,000 emails	Email (labeled)	CC BY 4.0
HuggingFace ealvaradob/phishing-dataset	HuggingFace Hub	~15,000 samples	Email + URL	MIT

Beyond labeled datasets, the following tools and plugins are available to augment the development pipeline:

- OpenPhish Feed: Real-time phishing URL intelligence feed, updated continuously, providing verified malicious URLs for streaming data augmentation.
- VirusTotal API: For URL and attachment reputation scoring, integrable into the inference pipeline as an external oracle.
- HuggingFace Transformers Library: Provides fine-tuning pipelines for models such as bert-base-uncased, roberta-base, and microsoft/deberta-v3-base.
- LangChain Framework: Enables prompt chaining and RAG (Retrieval Augmented Generation) integration for context-aware detection reasoning.
- spaCy & NLTK: For preprocessing, tokenization, and named entity recognition to extract key social engineering signals (urgency markers, authority claims).

3. Proposed Architecture & Methodology

3.1 System Architecture Overview

Our proposed system employs a multi-stage detection pipeline with three primary layers: (1) Preprocessing & Feature Extraction, (2) LLM-Based Semantic Classification, and (3) Ensemble Decision & Reporting.

Stage 1 — Preprocessing & Feature Extraction: Raw messages (email/SMS/chat) are ingested and decomposed into structural features (headers, DKIM/SPF validation results, sender reputation), URL features (domain age, redirect chains, entropy scores via VirusTotal API), and textual content normalized for tokenization.

Stage 2 — LLM-Based Semantic Classification: A fine-tuned RoBERTa-large model serves as the primary classifier for semantic analysis of message body content. This model is fine-tuned on a combined dataset of ~150,000 labeled samples drawn from the datasets listed in Section 2. A secondary LLM (LLaMA-3-8B via Ollama) is employed in a few-shot prompting configuration to provide explainable classification rationales for borderline cases.

Stage 3 — Ensemble Decision & Reporting: Outputs from the URL reputation scorer, structural feature classifier, and semantic LLM are fused via a logistic regression meta-classifier. The final output includes a binary verdict, confidence score, and a natural language explanation generated by the LLM module.

3.2 Fine-Tuning Protocol

- Base Model: roberta-large (HuggingFace Transformers)
- Training Data: 120,000 samples (60% phishing, 40% legitimate) with stratified train/val/test split (80/10/10)
- Optimization: AdamW optimizer, learning rate 2e-5, linear warmup schedule over 500 steps
- Hardware Target: Single NVIDIA A100 40GB GPU or equivalent (Google Colab Pro+)
- Epochs: 5 epochs with early stopping on validation F1 score
- Data Augmentation: Back-translation augmentation (English -> German -> English) for minority class samples using Helsinki-NLP/opus-mt models

3.3 Prompt Engineering Strategy

For the LLaMA-3 explainability module, we design a structured chain-of-thought prompt template:

```
System: You are a cybersecurity analyst specializing in phishing
detection. Analyze the following message and identify: (1) urgency
language indicators, (2) authority impersonation signals, (3) suspicious
URL patterns, (4) credential harvesting intent. Provide a structured
verdict with confidence level.
```

4. Evaluation Framework

4.1 Metrics

The system will be evaluated on the following quantitative metrics, aligned with operational security requirements:

Metric	Definition	Target Threshold
Precision	TP / (TP + FP)	>= 95% (minimize false positives)
Recall (Sensitivity)	TP / (TP + FN)	>= 92% (catch real phishing)
F1 Score	Harmonic mean of Precision & Recall	>= 93%
AUC-ROC	Area under ROC curve	>= 0.97
False Positive Rate	FP / (FP + TN)	<= 3% (operational usability)
Inference Latency	Average time per classification	<= 500ms (real-time feasibility)

4.2 Baseline Comparisons

Our system will be benchmarked against the following established baselines:

- Naive Bayes classifier trained on TF-IDF features (traditional spam filter baseline)
- Google Safe Browsing API (industry-standard URL filtering benchmark)
- BERT-base fine-tuned on SpamAssassin corpus (prior deep learning baseline)
- Zero-shot GPT-4o classification via API (state-of-the-art LLM without fine-tuning)

5. Implementation Roadmap

Phase	Timeline	Deliverables	Milestone
Phase 1: Data Curation	Weeks 1-2	Cleaned, merged dataset; EDA report; class balance analysis	Week 14 Checkpoint
Phase 2: Baseline Models	Week 2-3	Naive Bayes, BERT baseline; initial metrics	Week 14 Checkpoint
Phase 3: Fine-Tuning	Weeks 3-4	Fine-tuned RoBERTa; training curves; ablation study	Week 15 Checkpoint
Phase 4: Pipeline Integration	Week 4-5	Multi-stage pipeline; API integration; ensemble meta-classifier	Week 15 Checkpoint
Phase 5: Evaluation & Report	Week 5-6	Full evaluation report; PoC demo; final integrated document	Week 16 Final

5.1 Technology Stack

- Language: Python 3.11
- LLM Framework: HuggingFace Transformers 4.40+, PEFT (LoRA adapters for LLaMA-3)
- Orchestration: LangChain 0.2 for prompt chaining and RAG pipeline
- Training Infrastructure: Google Colab Pro+ (A100 GPU), Weights & Biases for experiment tracking
- API Integrations: VirusTotal API v3, OpenPhish Feed, HuggingFace Inference API
- Evaluation: scikit-learn, seaborn for metric computation and visualization
- Deployment: FastAPI REST endpoint for PoC demonstration

6. Ethical Considerations & Limitations

6.1 Ethical Considerations

This project operates exclusively with publicly available, ethically sourced datasets. No real user communications will be collected or processed without explicit consent. The system is designed for defensive detection purposes only, and no offensive capabilities will be developed. The team commits to responsible disclosure principles in the event that novel attack patterns are identified during research.

6.2 Known Limitations

- Adversarial Robustness: Fine-tuned models may be susceptible to adversarial prompt injection attacks specifically crafted to evade LLM-based classifiers. Future work should include adversarial training.
- Language Bias: Current datasets are predominantly English-language; performance on multilingual phishing content (e.g., Traditional Chinese phishing targeting Taiwan's financial sector) may be degraded.
- Concept Drift: Phishing tactics evolve rapidly; the model requires periodic retraining on fresh data to maintain operational effectiveness. A continuous learning pipeline is proposed as a future enhancement.
- False Positive Impact: Even at a 3% false positive rate, high-volume environments (e.g., enterprise email with 100,000+ daily messages) may generate thousands of incorrect flags, necessitating human-in-the-loop review workflows.
- Computational Cost: LLaMA-3 inference introduces latency overhead; production deployment would require quantization (GGUF/Q4_K_M) or dedicated GPU inference infrastructure.

6.3 Future Work

Upon completion of the PoC, several enhancements are identified for future research directions:

- Cross-lingual phishing detection using multilingual LLMs (mBERT, XLM-RoBERTa)
- Real-time streaming detection integrated with email gateway APIs (Microsoft Graph API, Gmail API)
- Federated learning framework to train on enterprise email data without centralizing sensitive communications
- Integration of behavioral biometrics signals (typing patterns, mouse dynamics) as additional social engineering detection features

7. Conclusion

This proposal presents a technically rigorous and operationally grounded approach to phishing and social engineering detection using fine-tuned LLM pipelines. The domain offers exceptional data availability — with over 650,000 labeled samples across email, URL, and SMS modalities accessible through public repositories — and a mature tooling ecosystem including HuggingFace

Transformers, LangChain, and VirusTotal API integrations. The multi-stage architecture combining semantic LLM classification with structural and URL-based feature analysis is designed to achieve detection performance exceeding current baselines while maintaining real-time inference feasibility.

The project scope is well-suited for the course timeline, with clearly defined milestones aligned to the Week 14, 15, and 16 checkpoints. The team is confident that this work will produce a publication-quality PoC system demonstrating the tangible value of LLM-powered cybersecurity tooling.

8. Work Distribution & AI Disclose

Project Founder— 徐祥智

Datasets & Available Resources— Claude

Reference Collection— Perplexity

Proposal Draft— Claude

Proofreading— 徐祥智

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